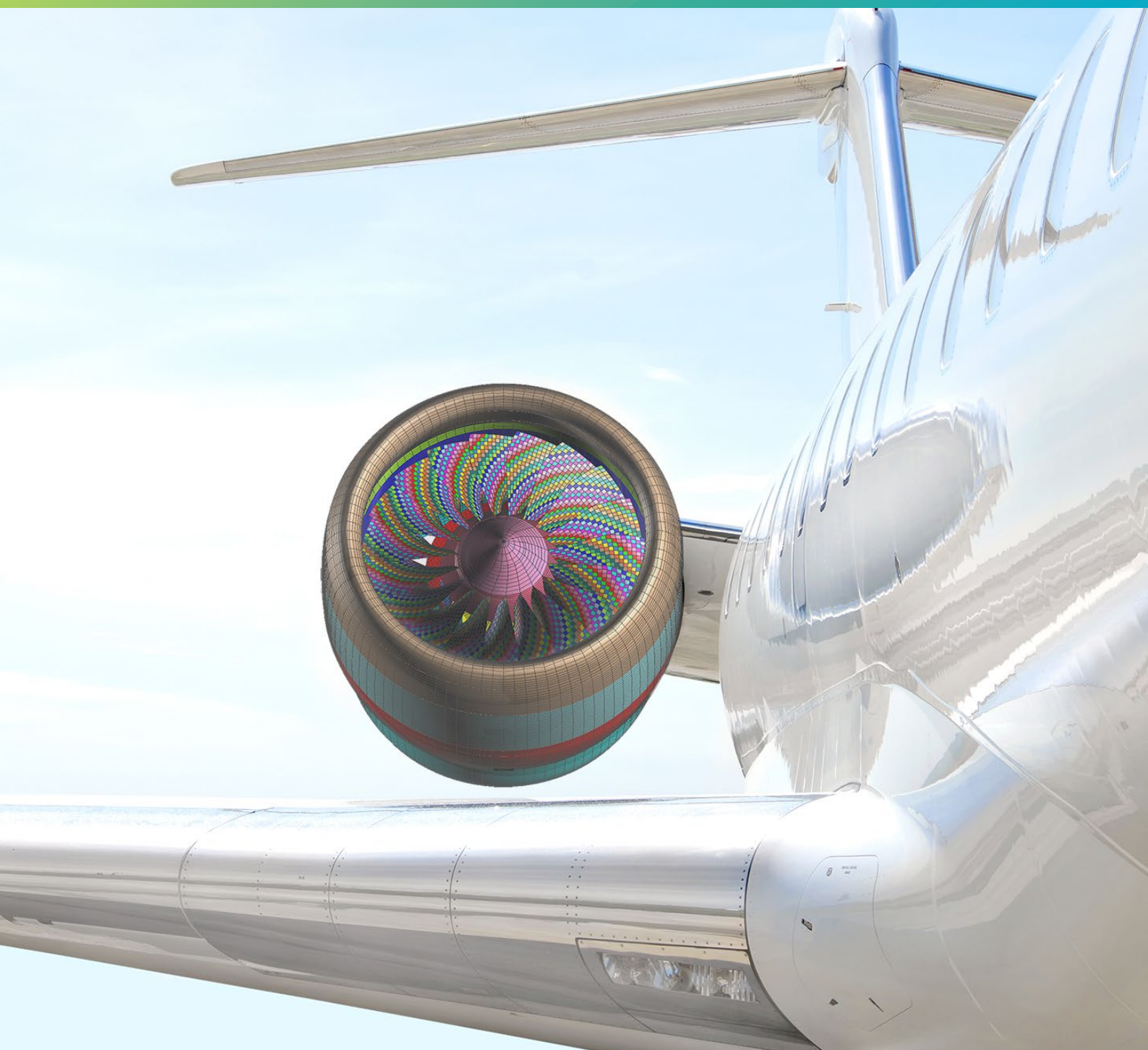


Virtual prototyping and engineering simulation provider to Aerospace and Defense industries

A partner for fast-growing Aerospace and Defense markets

Strengthening the Aerospace industry through engineering, innovation, and collaboration

MSC Software (MSC) is the leading virtual product development solution partner to the Aerospace and Defense Industry.



Through delivery of enhanced computer-aided virtual prototyping and test methods, simulation software, services, training, and support; product manufacturers and research institutes are developing the most innovative, high performing, and safe products in the world.

Together we strengthen the economy and improve local competitiveness

Delivering on time and in budget are the greatest challenges the Aerospace and Defense industry faces when developing products of increasing sophistication and complexity. Because development requires a broad and intricate conglomeration of private and government owned entities working together, including an extensive supplier network, efficiency is imperative.

MSC Software helps Aerospace and Defense companies stay innovative and competitive by partnering with and providing specialized simulation technology and expertise aimed at reducing costs and saving valuable time and effort throughout the life of critical projects and programs.

How MSC Helps local economy meet growth objectives for the Aerospace industry

Partnering with MSC Software helps strengthen the Aerospace Industry in the following ways:

- Improves industry competitiveness through use of proven simulation technology
- Increases engineering knowledge of virtual prototyping methods
- Improves engineering know-how through online eLearning courses and custom training
- Enables faster project completion through quick deployment of engineering software

Software deployment offerings:

Simulation components for R&D

- Domain technology transfer for R&D
- Fatigue & Fracture Mechanics Simulation: Marc
- Aero elasticity: MSC Nastran and toolkits
- Doublet Lattice method
- Hybrid Static Aero elasticity coupling
- Acoustics: Actran
- Rotor Dynamics: Nastran
- Materials idealization: Digimat



- Multi Body Dynamics Simulation: Adams
- Advanced Simulation Training and eLearning
- Comprehensive Support Offerings

Simulation lifecycle platform for engineering data management

Hosting of simulation data associated with multi domain simulation disciplines:

- Structural
- Thermal
- Acoustics
- Multi body and kinematics
- Systems and controls
- Engineering lifecycle management platform enrichment
- Materials from test to engineering domains
- Materials virtualization

Simulation lifecycle platform for aircraft

- Configuration Simulation Data Management (SDM) backbone into simulation data & process management (SDPM)
- Structural sizing
- Thermal sizing
- Landing gear simulation and test correlation
- Flaps simulation and test correlation
- Engine noise and transmittance to cabin
- SDPM extension to A/C architect trade off
- Integration of requirements and functional model management
- Trade off process interface deployment
- Kick off MoSSEC platform from SDPM
- Modeling simulation information in collaborative systems engineering context (sponsored by Airbus & Boeing along their ASD strategic standardization group engagement)

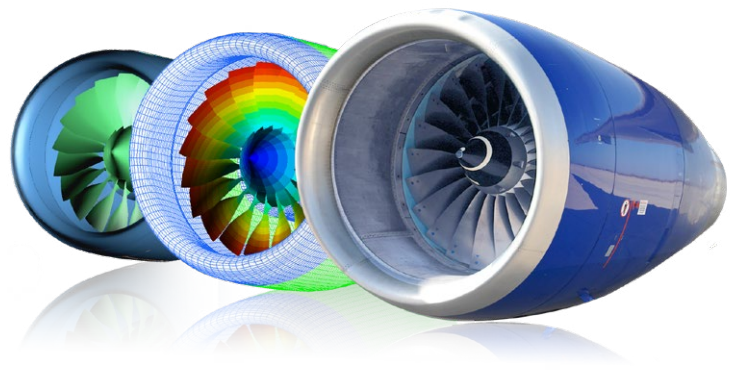
How to engage with MSC Software:

We provide a phased, easy, and quick approach to getting started on any project. It begins with a needs assessment and then transitions to training, software deployment, services and consulting, and support.

Depending on the goals of the project or program, we will work with you to customize an approach for working together. The phases below can be customized in any order to meet the unique needs of the client.

Project phases

- Needs/Process Assessment
- Review client internal development process from concept to production
- Initial assessment of virtual prototyping and test capabilities and needs
- Return on Investment (ROI) Model development
- Training / Theory & Practical Applications
- eLearning / virtual training
- Custom On-site courses



- Implementation and Deployment
- Computer Aided Engineering (CAE) Software for Research and Development
- Simulation Lifecycle for Engineering Data Management
- Simulation Lifecycle Platform for Aircraft Development and Virtual Test
- Technology Transfer
- Services
- Simulation projects
- Methods development
- Automation
- Validation and Verification
- Materials Testing
- Support (type of support)
- Mentoring
- On-site support
- Phone support

For more information, visit www.mscsoftware.com, or call your local MSC representative.



Hexagon is a global leader in digital reality solutions, combining sensor, software and autonomous technologies. We are putting data to work to boost efficiency, productivity, quality and safety across industrial, manufacturing, infrastructure, public sector, and mobility applications.

Our technologies are shaping production and people-related ecosystems to become increasingly connected and autonomous – ensuring a scalable, sustainable future.

Hexagon's Manufacturing Intelligence division provides solutions that use data from design and engineering, production and metrology to make manufacturing smarter. For more information, visit hexagonmi.com.

Learn more about Hexagon (Nasdaq Stockholm: HEXA B) at hexagon.com and follow us [@HexagonAB](https://twitter.com/HexagonAB).